

Liang-Ching Tao

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Research Interests

AI planning; structured sequential decision-making; reinforcement learning; learning-guided search; neural-guided Monte Carlo Tree Search; graph-based policy learning; deep neural representations; structured combinatorial optimization.

Education

National Chung Hsing University (NCHU), M.S. in Computer Science and Engineering 2026

Advisor: Prof. Pi-Chung Wang. Graduate GPA: 3.84/4.00.

Thesis: *A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.*

National Chengchi University (NCCU), B.S. in Computer Science 2023

Research Experience

Learning-Guided Multi-Charger Planning with Neural-Guided Search September 2026–present

- Developing a learning-guided planning framework for multi-charger scheduling that combines graph-based policy representations with neural-guided search over large, structured action spaces.
- Investigating canonical actions, action-space reduction, and sampled-action ideas for allocating limited search computation more effectively.
- Evaluating multi-agent coordination as a possible extension while keeping neural guidance for search, rather than a completed MARL system, as the current research focus.

Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging 2026–present

- Developed LP-BTS, a learning-guided planning architecture that combines a graph proposal policy and learned value critic with edge-budgeted PUCT for one-to-many mobile charging.
- Formulated scheduling over a large, dynamic, geometrically structured action space and investigated sampled candidate support for allocating bounded search computation.
- Designed matched ablations and a prospectively specified 30-scenario evaluation bank to assess end-to-end planning utility against learned and domain-engineered baselines.

Hierarchical RL for Multi-Charger Scheduling — M.S. thesis, NCHU 2025–2026

- Formulated coupled charging, routing, and allocation as a structured sequential decision problem in wireless rechargeable sensor networks.
- Developed a two-level decision architecture in which Q-learning selects regions or scheduling decisions and force-aware routing determines charger movement; built controlled evaluation workflows.

Research Assistant — Raman Spectral Analysis and Detection System, NCHU 2025–2026

- Developed an end-to-end applied deep-learning workflow spanning spectral preprocessing, learned 1D representations, binary decision models, and web integration.
- Processed .spc and .asc spectra through range restriction, fixed-length interpolation, baseline correction, smoothing, and intensity normalization; deployed residual 1D features with positional embeddings and multi-head attention for binary Sudan Red I–IV detection in a FastAPI/Next.js application. Repository

Research Experience (continued)

Research Assistant — Citrus Crop Disease and Pest Recognition, NCHU 2025–2026

- Developed a PyTorch image-classification workflow for citrus leaf diseases and pest damage; trained and compared EfficientNetV2-S and ResNet50 classifiers.
- Applied Focal Loss for class imbalance, MixUp/CutMix augmentation, and Grad-CAM inspection of lesion-focused evidence; exported trained models to ONNX for cross-platform integration. Inference service repository

Papers and Presentations

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- L.-C. Tao and P.-C. Wang, “A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks,” *2026 MCE&WASN Conference, National Chung Hsing University, Aug. 2026. Poster presentation.*
 - **Liang-Ching Tao**, Wei-Cheng Tai, Pi-Chung Wang. “Adaptive Charging Threshold and Detour Scheduling for Wireless Rechargeable Sensor Networks via Deep Reinforcement Learning.” *Accepted for oral presentation at MAMM 2026.*
 - **Liang-Ching Tao**, Pi-Chung Wang. “HQARF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation at MAMM 2026.* Preprint
 - **Liang-Ching Tao**, Po-Cheng Fu, Pi-Chung Wang. “DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation at MAMM 2026.*
 - **Liang-Ching Tao**, Wei-Chen Lin, Pi-Chung Wang. “CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation at MAMM 2026.*

Selected Research Methods

AI planning and sequential decision-making; learning-guided search and Monte Carlo Tree Search; reinforcement learning; graph-based policy representations; structured combinatorial optimization; neural representation learning; 1D spectral modeling; convolutional and attention-based architectures; experimental design, ablations, and reproducible simulation.

Teaching Experience

Teaching Assistant — Computer Networks (Undergraduate), NCHU 2025

- Supported Prof. Pi-Chung Wang through assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — Computer Networks (Professional Master’s Program), NCHU 2025

- Supported Prof. Pi-Chung Wang through assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.

Teaching Assistant — New Trends in Network Technology (Graduate Course), NCHU 2025

- Supported Prof. Pi-Chung Wang through assignment grading, student Q&A, lecture and laboratory assistance, and midterm/final-exam proctoring.